

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. We distinguish the deterministic hard-sphere flow, its Koopman derivation, law pushforward, ensemble log-Laplace functional, and limiting kinetic value. The exponential collision Hamiltonian is derived from the ensemble Green boundary jump, not from exponential conjugation of an interior derivation. An exact law-hierarchy intertwiner and a $1/j!$ -weighted full triangular corrector give generator convergence. The dynamic entropy action constructs a Nisio semigroup and discounted nonlinear resolvent; primal resolvent comparison avoids uncontrolled doubled jets. One-time microcanonical preparation and the full density-contact Gaussian risk-sensitive tangent are then obtained from B1–B3.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: ENSEMBLE BOUNDARY GENERATORS AND THE KINETIC NISIO SEMIGROUP

Five objects are kept distinct: the deterministic hard-sphere flow, its Koopman action on microscopic observables, the pushforward of probability

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laws, the ensemble log-Laplace functional, and the limiting kinetic Lax–Oleinik semigroup. A deterministic Koopman derivation is never claimed to create an exponential Hamiltonian. The exponential collision term arises from the boundary Green formula after the empirical jump is multiplied by the large-deviation speed.

1.1. The finite-volume boundary identity. Let Φ_t^ε be the hard-sphere flow, $U_t^\varepsilon G = G \circ \Phi_t^\varepsilon$, and $(\Phi_t^\varepsilon)_\# \rho$ the law pushforward. Let $X^\varepsilon = (\pi^\varepsilon, \Gamma^\varepsilon)$ be the empirical density/contact state. For a smooth density cylinder $F(\pi)$, write $p = \delta F / \delta \pi$.

Theorem 1.1 (Ensemble exponential Green identity). *If ρ_t is an ensemble law in the B2 trace graph, then*

$$\begin{aligned} \frac{d}{dt} \int e^{\mu_\varepsilon F(\pi^\varepsilon)} d\rho_t &= \mu_\varepsilon \int e^{\mu_\varepsilon F} \langle \pi^\varepsilon, v \cdot \nabla_x p \rangle d\rho_t \\ &\quad + \sum_{i < j} \int_{\Sigma_{ij}^-} \left[e^{\mu_\varepsilon F(\pi^{\varepsilon,+})} - e^{\mu_\varepsilon F(\pi^{\varepsilon,-})} \right] d\gamma_{\rho_t, ij}^-. \end{aligned}$$

At one contact,

$$\mu_\varepsilon (F(\pi^{\varepsilon,+}) - F(\pi^{\varepsilon,-})) = \Delta_\omega p + O(\mu_\varepsilon^{-1}).$$

Thus the ensemble logarithmic derivative contains $e^{\Delta_\omega p} - 1$. Pointwise in the regular interior, $A_\varepsilon e^{\mu F} = \mu e^{\mu F} A_\varepsilon F$, as required for a Koopman derivation.

Proof. Apply the B2 Green trace formula to the microscopic observable $e^{\mu_\varepsilon F(\pi^\varepsilon)}$. Free transport obeys the ordinary chain rule. Across a contact, the empirical measure changes by

$$\pi^{\varepsilon,+} - \pi^{\varepsilon,-} = \frac{1}{\mu_\varepsilon} (\delta_{v'} + \delta_{v'_*} - \delta_v - \delta_{v_*}),$$

so Taylor expansion gives the second formula. The boundary jump, not the interior derivation, is responsible for the exponential. $\square$

For a law ρ , define the finite ensemble Hamiltonian by dividing the right side by $\mu_\varepsilon \int e^{\mu_\varepsilon F} d\rho$. B2 trace convergence then gives the Hamilton–Boltzmann Hamiltonian.

1.2. Factorial hierarchy intertwining. Let g_j^ε be the factorial correlation hierarchy. A polynomial law cylinder has expansion

$$\mathcal{I}_\varepsilon F(g) = \sum_{j \geq 0} \frac{1}{j!} \langle f_j[F], g_j^\varepsilon \rangle.$$

The hierarchy norm is

$$\|g\|_{\alpha, \beta} = \sum_{j \geq 0} \frac{e^{\alpha j}}{j!} \|g_j\|_\beta.$$

Lemma 1.2 (Exact law–hierarchy intertwiner). *The map $\rho \mapsto (g_j^\varepsilon)_j$ intertwines the Liouville pushforward of laws with the BBGKY hierarchy, including the B2 normal-flux boundary operator. For every polynomial cylinder F , the ensemble Green identity equals the hierarchy duality*

$$\frac{d}{dt} \mathcal{I}_\varepsilon F(g_t) = \mathcal{I}_\varepsilon (\mathbb{A}_\varepsilon F)(g_t).$$

Proof. Expand the symmetric expectation in factorial moments. Free transport acts on each marginal. Integrating the contact boundary of a j -particle marginal gives the loss term, while summing over one unobserved label gives the $j+1 \rightarrow j$ creation term with its exact combinatorial coefficient. This is the BBGKY hierarchy. Absolute convergence follows from the $1/j!$ weight. $\square$

1.3. The full triangular corrector. Let F be a smooth kinetic cylinder. Expanding the finite ensemble Hamiltonian gives a triangular sequence of connected defects $D_j^\varepsilon[F]$. Let $\mathbb{A}_\varepsilon^{(J)}$ be the finite J -level observable hierarchy generator on its resolvent graph domain.

Theorem 1.3 (Normally summable hierarchy corrector). *For every regular moment ball and $T \leq T_0$, there are graph-domain correctors $c_j^\varepsilon[F]$ satisfying the backward equations*

$$\partial_t c_j^\varepsilon + \mathbb{A}_{\varepsilon,j} c_j^\varepsilon = -D_j^\varepsilon[F], \quad c_j^\varepsilon(T) = 0,$$

and

$$F_\varepsilon = F + \sum_{j \geq 2} \mu_\varepsilon^{1-j} c_j^\varepsilon[F].$$

Moreover,

$$\sum_{j \geq 2} \frac{e^{\alpha' j}}{j!} (\|c_j^\varepsilon\| + \|\mathbb{A}_{\varepsilon,j} c_j^\varepsilon\|) < \infty$$

uniformly in ε , and

$$\mathcal{H}_\varepsilon F_\varepsilon \longrightarrow \mathbb{H} \left(f, \frac{\delta F}{\delta f}, \psi \right)$$

locally uniformly.

Proof. Solve the finite triangular terminal system by the resolvent/Yosida propagator. B2 connected genealogies give

$$\|D_j^\varepsilon[F]\| \leq j! C^j T^{j-1}.$$

The $j!$ is exactly cancelled by the hierarchy norm, while the Boltzmann–Grad coefficient μ_ε^{1-j} and a loss $\alpha - \alpha' > CT$ make the series geometric. The same estimate holds for the generator images and time derivatives, so closedness of the graph domain passes the finite- J solutions to the infinite sum. Lemma 1.2 and the ensemble boundary identity identify the remaining binary term with the exponential collision Hamiltonian. B2 dominated genealogy convergence removes cycles. $\square$

1.4. The dynamic action and Nisio semigroup. For a balanced density/contact path on $[s, t]$, let

$$\mathcal{A}_{s,t}(f, \Gamma) = \int_s^t \ell(d\Gamma/dA_f) dA_f.$$

This transition action contains no initial preparation cost. Define

$$(S_{s,t}\Phi)(f_s) = \sup_{(f,\Gamma): f(s)=f_s} \{\Phi(f_t) - \mathcal{A}_{s,t}(f, \Gamma)\}.$$

Theorem 1.4 (Kinetic Nisio semigroup). *On every B2 compact action sublevel, $S_{s,t}$ is monotone, cash additive, a contraction in the sup norm, and*

$$S_{r,t} = S_{r,s}S_{s,t}.$$

For $\lambda > 0$, the discounted nonlinear resolvent

$$(J_\lambda h)(f) = \sup_{(f,\Gamma)} \int_0^\infty e^{-\lambda t} [h(f_t) - \ell(d\Gamma/dA_f)] dt$$

is well defined, satisfies $\|J_\lambda h - J_\lambda k\|_\infty \leq \lambda^{-1} \|h - k\|_\infty$, and is the resolvent of the Nisio generator associated with $\mathbb{H}$.

Proof. Concatenation of balanced paths preserves balance and adds the dynamic entropy, which proves both inequalities in the semigroup identity. Exponential discount and entropy coercivity make the infinite-horizon supremum finite. Splitting at a deterministic time gives the resolvent dynamic-programming equation. Subtracting two such equations proves the contraction estimate. On smooth cylinders, Fenchel duality for $\ell(q)$ gives

$$\lambda u - h = \mathbb{H}\left(f, \frac{\delta u}{\delta f}, 0\right)$$

in the graph-viscosity sense, identifying the generator. $\square$

1.5. Comparison without uncontrolled doubled jets. A bounded upper-semicontinuous u is a subsolution when $u \leq J_\lambda(h + \lambda u)$, and a bounded lower-semicontinuous v is a supersolution when $v \geq J_\lambda(h + \lambda v)$.

Theorem 1.5 (Primal resolvent comparison). *If u and v are respectively a sub- and supersolution of the same discounted equation, then $u \leq v$. The backward terminal equation*

$$\partial_s U + \mathbb{H}(f, \delta U / \delta f, 0) = 0, \quad U(T) = \Phi$$

has the unique bounded graph-viscosity solution $U(s) = S_{s,T}\Phi$. No doubled exponential jet is used.

Proof. Monotonicity and the resolvent contraction give

$$\|(u - v)^+\|_\infty \leq \|[J_\lambda(h + \lambda u) - J_\lambda(h + \lambda v)]^+\|_\infty \leq \|(u - v)^+\|_\infty.$$

Insert the strict perturbation $u_\eta = u - \eta$ in the subsolution inequality; the first inequality then improves by η/λ , forcing $(u_\eta - v)^+ = 0$. Let $\eta \downarrow 0$. Time discretization by the resolvent and the Crandall–Liggett product formula

gives the Nisio semigroup. Its dynamic-programming equation proves the terminal statement. $\square$

1.6. Microscopic convergence and preparation.

Theorem 1.6 (Ensemble-to-kinetic semigroup convergence). *For every bounded terminal cylinder and compact regular initial-law family, the finite ensemble log-Laplace values converge locally uniformly to $S_{s,t}\Phi$. Primitive microcanonical preparation is applied once at the initial boundary through the B1 source-dependent saddle. It is not reselected on subintervals.*

Proof. The corrected tests of Theorem 1.3 give upper and lower nonlinear-generator limits on a separating graph core. B2 exponential compact containment gives half-relaxed limits. The primal comparison theorem identifies both limits with the Nisio value. B1 conditions the initial law before the deterministic evolution; the fixed initial multiplier may be carried as a static coordinate but does not enter the dynamic action. $\square$

Theorem 1.7 (Gaussian risk-sensitive tangent). *Under the B3 central scaling, the kinetic semigroups converge on smooth finite-dimensional cylinders to the risk-sensitive diffusion semigroup with generator*

$$b \cdot DF + \frac{1}{2}A : D^2F + \frac{\theta}{2}DF^T ADF,$$

where A is the full density/contact covariance, including pure-contact noise and the B1 initial Schur complement.

Proof. B3 gives process convergence and a uniform complex moment neighborhood. Taylor expansion of the finite ensemble pressure shows that cumulants of order at least three vanish under the central scaling, while the first two give b and A . Uniform integrability passes the exponential transform to the limit. Applying the logarithmic transform to the diffusion generator yields the quadratic term. $\square$

Theorem 1.8 (Round-fourteen B4 closure). *The deterministic hard-sphere ensemble has a correctly typed boundary nonlinear generator, a normally summable hierarchy corrector, a constructed kinetic Nisio semigroup with primal comparison, a one-time microcanonical interface, and the full Gaussian risk-sensitive tangent.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.5, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.